TIEN – ANH NGUYEN

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Summary

Materials Science Ph.D. Candidate with extensive expertise in Thin Film Deposition (PLD, CVD, Sputtering), Device Fabrication (E-beam lithography, Photolithography, Dry-etching), Material Characterization (thin-film XRD, SEM, AFM, XPS), and Electrical Characterization (Keithley 4200A-SCS with SMU, CVU, PMU). Proficient in combining DFT simulations (VASP) with experimental fabrication to accelerate device mechanisms and development. Pioneered the first-ever integration of single-crystalline entropy-stabilized oxide (ESO) memristors onto Silicon substrates, successfully engineering a Back-End-of-Line (BEOL) compatible process with a thermal budget under 450°C. Authored 7 peer-reviewed publications in high-impact journals such as ACS Applied Materials & Interfaces, Materials Today Energy, and Ceramics International.

Education

Ph.D. in Materials Science, Oregon State University, Corvallis, Oregon — Expected May. 2028

M.Sc. in Nano Devices and System, Inje University, South Korea — Feb. 2023

B.Eng. (Hons.) in Engineering Physics, Hanoi University of Science and Technology, Vietnam — Jan. 2021

Skills

- **Electrical characterization** — Semiconductor analysis (Keithley 4200A-SCS), DC I-V sweeps (SMU), C-V profiling (CVU), ultra-fast transient Pulse measurements (PMU), Field-Effect Transistor (FET) characterization (I_{DS}-V_{GS}, I_{DS}-V_{DS}), Four-Point Probe, Cyclic Voltammetry (CV), Electrochemical Impedance Spectroscopy (EIS), Battery Cycling.
- **Device fabrication** — Photolithography, E-beam lithography, Laser lithography, Mask aligner, Spin Coating, Spray Coating, Dry etching, Wet etching, Plasma Treatment.
- **Computing Codes** — Vienna Ab initio Simulation Package (VASP), Alloy Theoretical Automated Toolkit (ATAT), VASPKIT, LOBSTER, Python.
- **Material synthesis and characterization** — Pulsed Laser Deposition (PLD), Sputtering, Chemical Vapor Deposition (CVD), Thermal Evaporator, Electrochemical Deposition.
  Scanning Electron Microscopy (SEM), Transmission Electron Microscopy (TEM), X-ray Photoelectron Spectroscopy (XPS), powder and thin-film X-Ray Diffraction (XRD), Atomic Force Microscopy (AFM), Fourier Transform Infrared spectroscopy (FTIR), UV-Vis Spectroscopy, Raman Spectroscopy, Contact Angle.

Experience

Research Assistant, Chae Lab and Brain-inspired Computing Lab, OSU, Corvallis, OR — Sept. 2024 – present

1. Entropy-Stabilized Oxide (ESO) Memristors for BEOL Integration

- Characterized complex device switching kinetics utilizing specialized semiconductor characterization systems (Keithley 4200A-SCS equipped with SMU, CVU, and PMU modules) to evaluate short-term memory dynamics.
- Spearheaded the first-ever integration of single-crystalline Entropy-Stabilized Oxide (ESO) memristors on Si substrates via Pulsed Laser Deposition (PLD) using an Ion-Beam Assisted Deposition (IBAD) MgO buffer layer. Engineered a Back-End-of-Line (BEOL) compatible fabrication flow by maintaining a thermal budget of < 450 °C.
- Demonstrated device performance on Si equivalent to films grown on bulk MgO substrates, exhibiting robust short-term memory dynamics for neuromorphic computing. Validated industry-leading reliability with a switching endurance of 10^9 cycles.

2. Study atomistic mechanism and improve performance of Selector-Only-Memory (SOM)

- Engineered an environmentally sustainable, Bismuth (Bi)-doped GeSe chalcogenide to replace conventional Arsenic (As)-doped materials while maintaining high-performance.
- Conducted first-principles Density Functional Theory (DFT) calculations using VASP to investigate the atomistic mechanisms of the amorphous network.
- Correlated theoretical models with experimental device measurements, demonstrating optimized threshold voltage () and leakage current profiles suitable for high-density memory arrays.

3. RRAM-Based Synaptic Devices for Neuromorphic Computing

- Fabricated and characterized TaO_x/Ta_2O5 and $SrCoO_x$ RRAM-based synaptic devices (up to array level) for neuromorphic computing.

- Optimized photolithography and etching parameters to improve device-to-device (D2D) and cycle-to-cycle (C2C) uniformity, critical for large-scale neuromorphic integration.
- Characterized device switching kinetics using semiconductor analyzer (Keithley 4200A-SCS)

Materials Engineer Intern, VINFAST, Vietnam Apr. 2024 – Jul. 2024

- Conducted systematic screening and performance benchmarking of anode materials for Electric Vehicle (EV) Li-ion batteries, optimizing for fast-charging kinetics and extended cycle life.
- Fabricated and assembled coin-cell test vehicles to characterize intrinsic material properties using advanced electrochemical techniques.
- Performed rigorous data analysis on battery cycling performance to identify degradation mechanisms, ensuring material specifications met strict industrial quality standards

Scientific Researcher, LAMP4PPM Lab, Sungkyunkwan University, South Korea Mar. 2023 – Mar. 2024

1. Li-ion Batteries recycling

- Conducted slurry formation, electrode fabrication, characterization, and electrochemical measurement.
- Investigated the impact of chemical impurities on the electrochemical behavior of recycled Li-ion batteries to establish critical tolerance limits.
- Determined precise impurity thresholds to enable cost-effective optimization of recycling workflows without compromising device reliability.

2. Roll-to-roll (R2R)-printed single-walled carbon nanotube-based thin film transistors

- Conducted comprehensive electrical measurements on R2R-printed thin-film transistors (TFTs).
- Extracted and analyzed critical electrical parameters, including field-effect mobility, threshold voltage (Vth), subthreshold swing (SS), and ON/OFF current ratios to benchmark manufacturing process variations.
- Optimized Roll-to-Roll (R2R) printing processes for single-walled carbon nanotube-based thin-film transistors (TFTs).
- Significantly enhanced transistor performance and carrier mobility through targeted surface treatments and interface engineering.

Research Assistant, Hyun Chul Kim's Lab, Inje University, South Korea Mar. 2021 – Feb. 2023

- Synthesized high-aspect-ratio Copper Nanowires (CuNWs), and Silver Nanowires (AgNWs) via a solution-phase polyol method, precisely controlling temperature and precursor concentration to ensure uniform nanowire dimensions.
- Optimized the addition of capping agents (e.g., PVP) to regulate the crystal growth direction, achieving high optical transparency and electrical conductivity for flexible transparent electrodes
- Developed a subsequent Ni-coating process onto the synthesized AgNWs to create a core-shell structure, significantly improving the material's chemical stability and non-enzymatic sensing performance.
- Developed flexible and transparent glucose sensors utilizing Ni-coated Silver Nanowires (AgNWs) networks on PET substrates.

Publications (Selected, [Google scholar]())

[1] **Nguyen, T.A.**, Han, C., Ghosh, D., Chae, S.Y., Lee, S.H., Chae, S. Determining the relationship between composition, structure, and device properties of Ge_xSe_{1-x}-based selector-only memory. Advanced Electronic Materials (In revision)

[2] Huang, Y.C., Lee C. W., **Nguyen, T.A.**, Kioupaskis. E., Chae, S. *(2025) Ferroelectricity of Wurtzite Be1-xMgxO From First-Principles Calculation.* *Applied Physics Letters, 127(14).*

[3] Le, X. P., Venkatesan, A., Daw, D., **Nguyen, T. A.**, Baithi, M., Bouzid, H., Nguyen, T. D. *(2024). High Performance p-type Quasi-Ohmic of WSe2 Transistors using Vanadium Doped WSe2 as Intermedia Layer Contact. ACS Appl. Mater. Interfaces 2024, 16, 39, 52645–52652.*

[4] **Nguyen, T. A.**, Nguyen, C. T., & Kim, H. C. *(2023). A flexible, transparent, and enzyme-free sensor based on Ni@ AgNW networks for glucose detection. Ceramics International, 49(11), 17611-17619.*

[5] **Nguyen, T. A.**, Luu, T. L. A., Do, D. T., Dang, D. V., Nguyen, H. L., Kim, H. C., Nguyen, C. T. *(2022). Photocatalytic, electrochemical, and electrochromic properties of in situ Ag-decorated WO_3 nanocuboids synthesized via facile hydrothermal method. Applied Physics A, 128(12), 1047.*

Conferences & Activities

- **Oral presentation:** “Determining the Relationship Between Composition, Structure and Device Properties of Ge_xSe_{1-x}-Based Selector-Only Memory” 2025 MRS Fall Meeting & Exhibit.
- **Participant:** Summer School for Integrated Computational Materials Education, University of Michigan, June 9 – 20, 2025.
- **Poster presentation:** "Improving the performance of R2R printed single-walled carbon nanotube-based thin film transistors through interface engineering” Korea Flexible & Printed Electronics Society Meeting, Busan, South Korea, March 27-29, 2024.